

Assignment 2.4: Connecting and Testing Peripherals

Task 1: Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

SET UP:



RESULT:

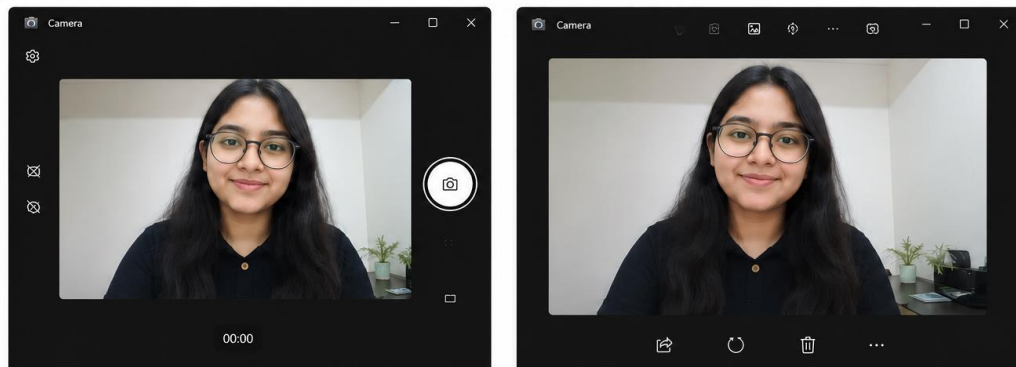


Fig 1. Web Cam Set up and its result

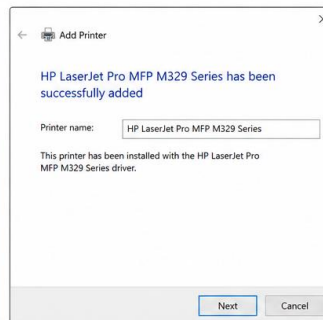
1. Web Cam Set-up
 - Connected the USB webcam to an available USB port on the computer
 - Windows automatically detected the webcam and installed the required driver.
 - Waited until the device was ready for use.
2. Driver Installation
 - The webcam was detected successfully by Windows.
 - The required USB webcam driver was installed automatically.
 - No additional driver installation was required.
 - The webcam appeared correctly in the system without any errors.
3. Verification
 - Opened the Windows Camera Application.
 - Confirmed that the webcam displayed a live video preview.
 - Captured a photo using the camera to verify proper functionality.
4. Result
 - The USB webcam was installed successfully.
 - The driver was installed automatically.
 - The webcam, worked correctly, and a photo was captured successfully.

Task 2: Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

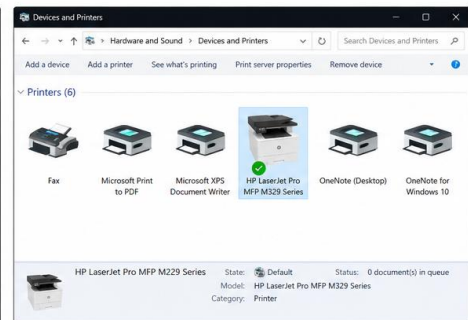
SET UP:



1. Printer connected using USB cable and powered on.

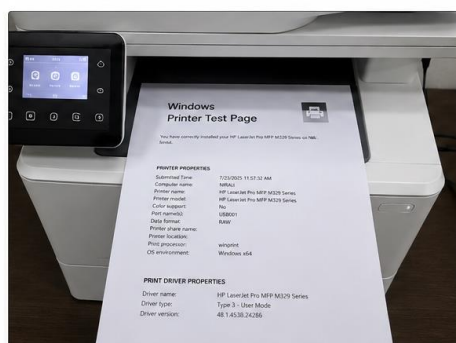


2. Correct driver installed successfully.

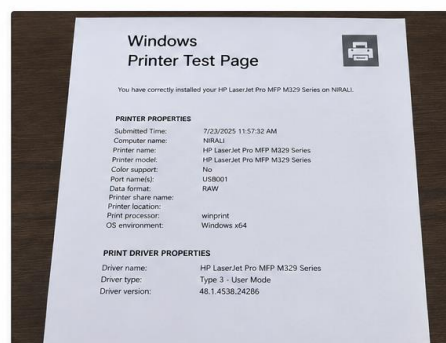


3. Printer listed in Devices and Printers.

RESULT:



4. Test page printing...



5. Test page printed successfully.

Fig 2. USB Printer Set up and Result

1. Printer Setup

- Connected the HP LaserJet Pro MFP M329 Series printer to the computer using a USB cable.
- Powered on the printer and ensured it was ready for use.
- Windows detected the connected printer automatically.

2. Driver Installation

- **Opened Settings > Bluetooth & devices > Printers & Scanners (or Control Panel > Devices and Printers).**
- Added the HP LaserJet Pro MFP M329 Series printer.
- Windows installed the required printer driver automatically (or the correct HP driver was selected if prompted).
- Verified that the printer status showed **Ready** after installation.

3. Test Page Verification

- Opened the printer properties.
- Selected Print Test Page.
- The printer printed the windows test page successfully, confirming that the installation was complete the printer was functioning correctly.

4. Result

- The HP LaserJet Pro MFP M329 Series printer installed successfully.
- The correct printer driver was installed.
- The printer was detected by Windows without any errors.

- A test page was printed successfully, confirming proper installation and operation.

Task 3: Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem. Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

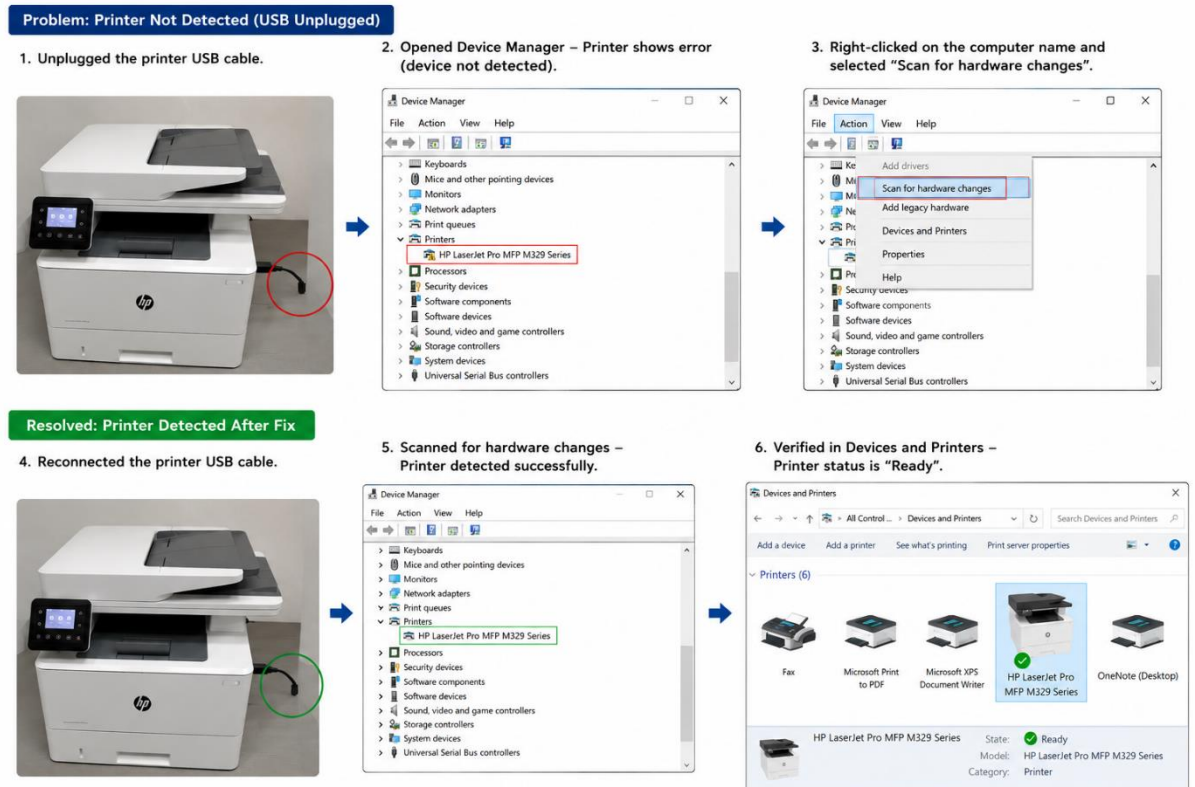


Fig 3. Printer Troubleshooting

1. Simulating the connectivity Issue

- Switched on the printer and ensured it was working properly.
- Unplugged the USB Cable connecting the printer to the computer.
- The computer could no longer detect the printer, and the printer became unavailable.

2. Identifying the problem

- Opened Device Manager by pressing Windows + X and selecting Device Manager.
- Checked the Printers or Print queues section.
- Observed that the printer was either missing or displayed a warning icon, indicating a connectivity issue.

3. Resolving the Issue

- Reconnected the USB cable securely to the printer and computer.
- In Device **Manager**, clicked **Action -> Scan for hardware changes**.
- Windows detected the printer again and reinstalled device automatically if required.

- Opened **Device and Printers** and verified that the printer status changed to **Ready**.

4. Result

- The printer was detected successfully after reconnecting the USB cable.
- **Scan for hardware changes** refreshed the device list and restored the printer.
- The warning/error disappeared, and the printer was available for use again.

Task 4: A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications. Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

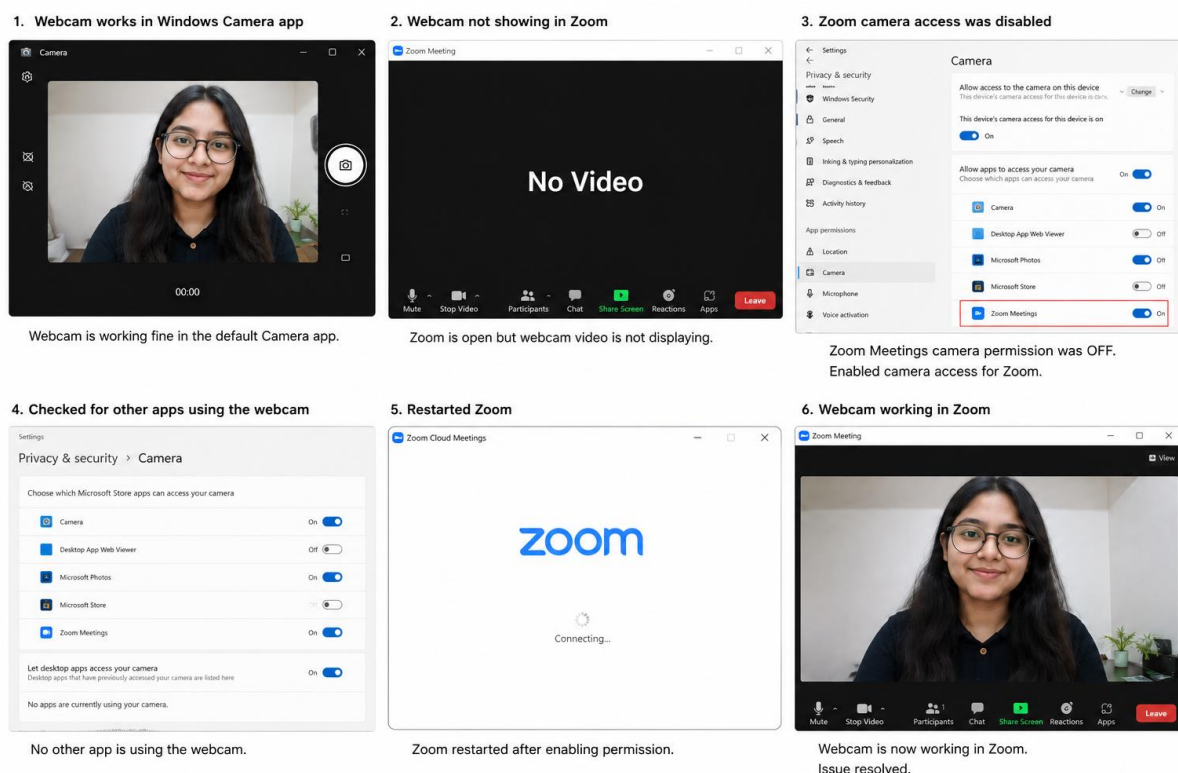


Fig 4. Zoom webcam Troubleshooting

1. Identify the issue

- Opened the Windows Camera application
- Verified that the webcam displayed a live video feed.
- Opened Zoom and noticed that the camera showed a black screen or “No Video”.

2. Troubleshooting Steps

- Checked Windows Camera Privacy Settings by having to Settings -> Privacy & Security -> Camera.
- Enable Camera access and Let desktop apps access your camera.
- Confirmed that Zoom had permission to use the webcam

- Closed any other applications that might be using the webcam
- Restarted Zoom and selected the correct webcam from Zoom Settings -> Video
- Tested the webcam again in a Zoom meeting.

3. Result

- Windows Camera application continued to work correctly.
- Camera permissions for Zoom were enabled.
- Zoom successfully detected the webcam.
- Live video was displayed correctly during the Zoom meeting